

Yash Verma

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EDUCATION

- **Indian Institute of Information Technology, Lucknow** Lucknow
Bachelor of Information Technology; CGPA: 8.32 Apr 2020 - Apr 2024
- **Relevant Courses:** Data Structures & Algorithms, Operating System, System Design, DBMS, Object Oriented Programming, Distributed Systems, Software Management, Design Patterns.

EXPERIENCE

- **Zus (formerly 0chain)** Cupertino, CA (Remote)
Software Development Engineer (Freelance) - Golang, GitHub Actions, Gin, Grafana, Helm, Kafka Apr 2024 - Present
 - Spearheaded the **UAT test suite** for **enterprise storage**, reducing production bugs by **90%** and significantly improving release quality.
 - Refactored and migrated **GoSDK**, reducing app size by **50%**, while assisting cross-platform teams in the transition.
 - Benchmarked **S3-compatible storage** for **500 million** bulk files and **40 TB** large files, achieving sub-**100ms** latency on both **Enterprise** and **Standard** allocations.
 - **Load-tested** streaming videos using **JMeter**, optimizing performance based on varying data sets and **Erasure Coding (EC) parity nodes** for efficient allocation.
 - Developed **automated workflows** for **view change** and dynamic removal of **Sharders/Miners**, boosting developer efficiency by over **90%**.
 - Built an **automated alert system** for **missing network blocks** and integrated **build tags** into Slack using **Python scripts**, **BeautifulSoup**, **Webhooks**, and **Block Kit**.
 - Enhanced **multi-wallet support** to improve **concurrency handling** in **load tests**, while refactoring legacy load test code with a unified **GoSDK** for streamlined performance.
- **Zus (formerly 0chain)** Cupertino, CA (Remote)
Android Engineer (Freelance) - Kotlin, Java, AndroidSDK, GitHub Actions Feb 2022 - Mar 2024
 - Architected two Android apps, **Bolt** (crypto wallet) and **Vult** (decentralized storage platform), increasing the Android user base by **20%** for decentralized storage using **ZCN crypto**.
 - Led and mentored two junior developers, enhancing **code quality** by **40%** and ensuring **95%** on-time delivery through effective **Scrum leadership**.
 - Designed and implemented **release pipelines** for **Android apps**, reducing manual effort and accelerating release cycles by **70%**.
 - Developed an **Android app** with **Split Key Functionality** to act as an intermediary between the **app** and the **blockchain**, enabling double-signing of transactions via **Bluetooth**.
- **Google Summer of Code Oppia Foundation** Remote
GSOC - Kotlin, GitHub Actions, JUnit, UIAutomator, Gradle, Bazel, AndroidSDK Jun 2021 - Sept 2022
 - Designed and architected a robust **CI/CD infrastructure** to streamline build processes and enhance deployment efficiency.
 - Implemented automated build triggers for **alpha**, **beta**, and **test** releases, seamlessly signing and uploading binaries to **Google Drive**, **GCP Bucket**, and **Google Play Store** tracks.
 - Developed an intelligent **cherry-picking workflow** that autonomously merges selected changes into the appropriate **flavor branch**.
 - Engineered an automated logging system that tracks each build and release step, updating **GitHub issues** with workflow triggers and contextual comments.
 - Spearheaded **Android Release Automation**, integrating streamlined processes for **signing**, **building**, **testing**, and **publishing** applications to Google Play.
 - Authored custom **Kotlin scripts** to simplify essential tasks such as generating **AAB bundles**, securely **signing** artifacts, and automating **Google Play** uploads.
 - Leveraged **GitHub Actions** in conjunction with custom **Kotlin scripts** executed via **Bazel** to optimize **CD pipelines** for **alpha**, **beta**, and **nightly releases** on scheduled **cron jobs**.
- **Oppia Foundation Github Link [Link to Merged PRs]** Mountain View, CA (Remote)
Open Source Contributor Android June 2021 - April 2022
 - Contributed to the migration of the build system from **Gradle** to **Bazel**, achieving a **10x** improvement in build speeds.
 - Developed modular **fragments** encapsulated within standalone **activities**, ensuring seamless functionality across mobile and tablet devices with support for **multiple screen sizes**.
 - Implemented **MVVM** (Model-View-ViewModel) and **MVP** (Model-View-Presenter) architectures to structure and optimize all activities and fragments.
 - Authored over **70+** Unit Tests, **20+** Instrumentation, and End-to-End tests using **Espresso**, **Robolectric**, and **UIAutomator**, with **JUnit** as the test runner for Android.
 - Mentored and onboarded **3+** students in the **Android repository**, guiding them to become active **collaborators**.
 - Enhanced project documentation by creating a comprehensive **GitHub Wiki** on **Protobuf**, significantly improving understanding and adoption of the **Protobuf** library within the repository.
- **Gnome Foundation** Remote
Open Source Contributor - Tech Stack: C, XML, GTK+ Sept 2021 - Feb 2022
 - Added a new feature in the **path-bar** of **Nautilus file manager**, improving **user efficiency** by **20%** by enabling seamless opening of folders and files.
 - Designed and implemented a new button using **XML**, enhancing **user interaction** by **15%** through improved interface functionality.

- Refactored and optimized **UI components** in Nautilus, reducing render time by **30%** and improving overall navigation fluidity.
- Implemented keyboard shortcuts for **quick folder navigation**, enhancing accessibility and workflow efficiency.
- Integrated **drag-and-drop support** for seamless file organization, significantly improving **user experience** in Nautilus.

BSHAPP

Andhra Pradesh

May 2021 – Jun 2021

Android Freelance Developer

- Added support for **multiple languages**, expanding the app's user base by **30%** and enhancing accessibility in diverse markets.
- Implemented **Jetpack Navigation** with single activity and multi-fragment architecture, improving app performance by **25%**.
- Integrated **REST API** with **Retrofit**, reducing data retrieval time by **20%** and improving HTTP request efficiency.
- Optimized **network caching** mechanisms using **OkHttp**, significantly reducing redundant API calls and enhancing data load speeds.
- Refactored legacy codebase to adopt **Kotlin Coroutines** for asynchronous processing, leading to a **40%** improvement in app responsiveness.
- Enhanced security by implementing **encrypted shared preferences** and **biometric authentication**, strengthening data protection and user privacy.

PROJECTS

- **Parking Lot Management System:** Developed a RESTful API using **Node.js**, **Express.js**, and **MongoDB** to manage parking spaces. Integrated **Winston** for logging and conducted unit and integration testing with **Jest**.
- **Airline ticketing system:** Built using **Spring Boot** in **MVC architecture** where a user can book tickets, query flights and cancel tickets also **Dockerized**.
- **Trip Feedback System:** Designed a feedback system for **Travelers** and **Transport Managers** with secure feedback submission, user authentication, using **Spring Framework**.
- **Festify:** Hackathon and event management multi-module Android app with NodeJs backend and MongoDB database.
- **Hello World:** Android application showcasing 0Chain blockchain storage SDK, designed in Kotlin.
- **Wake-Lock:** Runs a sticky service in the background which adds a partial wake lock to your screen.
- **Automated Window Manager:** A CLI tool to automate window management in Linux.
- **Backup System Ubuntu:** A bash script to back up apt, snap, and other config files.
- **GUI-Automator:** Python script for downloading, sorting, and extracting files from Google Drive.
- **What's App Clone:** Android chat app featuring Google and email authentication.
- **Video Chatting:** Android app for video conferencing using Jitsi Meet API with a secure server setup.
- **My Resume:** **Automated deployment** of resume updates using **GitHub Actions** and **GCP** to Google Drive.

ACHIEVEMENTS

- Practised over **700+** DSA problems on Leetcode.
- Member of **Oppia Foundation** for **2 years**.
- Completed **Cloud Engineering tracks** 30 Days of Google Cloud Program.

TECHNICAL SKILLS

- **Programming & Scripting:** Java, Go, Kotlin, Python, JavaScript, SQL, Bash, C/C++
- **Backend Frameworks:** Spring Boot, Gin, Express.js, REST/GraphQL APIs, RxJava, Dagger/Hilt
- **DevOps & Cloud:** Docker, Kubernetes, GitHub Actions, CI/CD, GCP, Linux, Helm, Prometheus/Grafana/Loki
- **Databases & Testing:** MySQL, MongoDB, PostgreSQL, Firebase, JUnit, Mockito, UIAutomator
- **Tools & Libraries:** Git, Retrofit/OkHttp, Gradle, Bazel, Android SDK (cross-platform support)

SOFT SKILLS

- **Communication:** Excellent verbal and written communication skills, effective team collaboration.
- **Leadership:** Strong leadership skills, mentoring and guiding junior developers.
- **Problem Solving:** Analytical and critical thinking, adept at problem-solving and debugging.
- **Adaptability:** Ability to adapt to new technologies and work in a fast-paced environment.

HOBBIES

- **Open Source:** Contributing to open-source projects, mentoring new developers.
- **Programming:** Solving DSA problems on Leetcode, Codeforces, and Codechef.
- **Reading:** Reading tech blogs, articles, and books on software development.
- **Music:** Playing guitar, listening to music, and attending live concerts.
- **Electronics:** Building electronic circuits, transformers, etc.